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PAPER

TF-crossnet: a cross-modal attention fusion network for cardiovascular disease classification using pcg and ecg signals

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Keywords: electrocardiogram, phonocardiogram, cardiovascular disease, multi-modality

Abstract

Electrocardiogram (ECG) and phonocardiogram (PCG) have emerged as crucial non-invasive and portable diagnostic modalities for early cardiovascular disease (CVD) screening. Despite the individual merits of these signal modalities in CVD detection, significant challenges persist, including insufficient inter-modal interaction and suboptimal weight allocation. To address these critical limitations, we proposed a novel Time-Frequency Cross-Modal Attention Fusion Network (TF-CrossNet) designed for precise early CVD diagnosis. The proposed network employs a dual-path multiscale residual structure to extract key time-frequency domain features from PCG and ECG signals, comprehensively capturing multiscale information. Leveraging the intrinsic electro-mechanical coupling relationship of the heart, a bidirectional mutual enhancement attention module is introduced to capture interactive morphological information between PCG and ECG signals, enabling feature-level signal complementation and enhancement. Furthermore, an adaptive fusion strategy based on Bayesian decision theory is developed, establishing a mapping relationship between confidence levels and loss functions to dynamically optimize modal weight allocation. Validated on the 2016 PhysioNet/CinC dataset, the model achieved exceptional performance metrics: 93.13% accuracy, 97.7% specificity, and 98% area under the curve (AUC). Furthermore, comprehensive noise robustness experiments demonstrate that TF-CrossNet maintains superior performance under various noise conditions, achieving an average robustness index of 94.20% compared to existing methods, validating its practical applicability in clinical environments. The superior effectiveness of the proposed approach in CVD classification, providing a novel technological pathway for non-invasive and precision CVD diagnosis.

1. Introduction

Cardiovascular disease (CVD), as the leading cause of death worldwide, causes about 17.9 million deaths annually (Zhang *et al* 2024a). In recent years, multi-modality fusion of electrocardiogram (ECG) and phonocardiogram (PCG) signals has attracted much attention due to its ability to provide information about the electrical and mechanical activities of the heart, respectively. The two types of signals belong to different modalities and are complementary (Ameen *et al* 2024), and their fusion has shown significant advantages in the diagnosis of CVD, which can significantly improve the diagnostic performance (Wang *et al* 2025).

Current automated diagnostic research mainly focuses on single-modality analysis, ECG is mainly used for arrhythmia and myocardial infarction detection (Hannun *et al* 2019) (Srivastava *et al* 2022) (Sun *et al* 2023), while PCG is used for valve disease diagnosis (Tuncer *et al* 2021) (Ghosh *et al* 2020). In recent years, multi-modality fusion technology has gradually become a research hotspot in the field of CVD automatic diagnosis (Huang *et al* 2024) (Tao and Velasquez 2022) (Hangaragi *et al* 2025) (Morshed and Fattah 2023). LI *et al* (2019) proposed for the first time a neural network framework based on dual inputs of ECG and PCG, which combines a convolutional neural network (CNN) with a bi-directional gated recurrent unit (GRU) through multidomain feature extraction (time and frequency domains, etc) and feature downscaling with an attentional mechanism, validating the effectiveness of multi-modality fusion in coronary artery disease (CAD) detection. Chakir *et al* (2020) extracted 10 biomarkers including PCG amplitude, RR interval variation, etc based on synchronized ECG and PCG signals, further demonstrating the potential of multi-modality fusion in enhancing diagnostic sensitivity. Li *et al* (2021) proposed a multi-input CNN framework combining a one-dimensional convolutional neural network (1D-CNN) and a two-dimensional convolutional neural network (2D-CNN) , which processed the original signals, Gram's angle fields (GAFs), and temporal images (S-transforms, MFCCs), respectively, to achieve automated feature extraction and classification. LI *et al* (2022) proposed a multi-modality cardiac function signal classification algorithm based on improved dempster-shafer (D-S) evidence theory (Zhang *et al* 2021). The feature vectors of synchronously acquired PCG and ECG signals were extracted by wavelet scattering transform, and these feature vectors were classified by using support vector machine (SVM) (Majeed and Alkhafaji, 2023). (Zhu *et al* (2025) proposed a dual-scale deep residual network (DDR-Net), which extracts features automatically from the original PCG and ECG signals, respectively, and employs a dual-scale feature aggregation module to integrate the underlying features at different scales (Huo *et al* 2024). Then support vector machine-recursive feature elimination with cross-validation (SVM-RFECV) was used to select important features and SVM was used for final classification. The multi-modality fusion technique of ECG and PCG provides a new research direction for the automatic diagnosis of CVD, which demonstrates significant advantages in feature extraction, feature fusion, and decision optimization, and is expected to become an important technological tool for CVD diagnosis in the future (Xue *et al* 2016) (Amal *et al* 2022).

In recent years, cardiovascular disease detection research based on ECG and PCG still faces key challenges. Existing methods fail to fully consider the differential dependence of CVD on the two modalities resulting in a lack of disease specificity in modality weight assignment, Additionally, most current studies use independent feature extraction strategies that fail to effectively model the inherent electro-mechanical coupling and temporal dependence between PCG and ECG. This neglect of multi-modality interaction mechanisms with simple feature-level fusion strategies limits the accuracy and robustness of multi-modality fusion in CVD diagnosis.

To address the above problems, this paper proposes a cross-modal attention fusion network (TF-CrossNet) based on time-frequency information, aiming to address the limitations of existing methods in multi-modality signal processing through innovative feature extraction and fusion strategies. The network realizes deep inter-modal feature interaction and complementary enhancement by extracting the time-frequency domain features of PCG and ECG signals respectively, combined with a bidirectional mutual enhancement attention module. In addition, the network introduces a probabilistic adaptive fusion strategy to integrate the confidence features of different modalities, which significantly improves the classification performance of the diagnostic model. The main contributions of this paper can be summarized as follows:

(1) The TF-CrossNet proposed in this paper was able to capture multi-scale time domain features and complex frequency domain information simultaneously.

(2) Based on the cardiac electro-mechanical coupling physiological relationship, a bidirectional mutual enhancement attention module was introduced to realize inter-modal feature complementation and enhancement.

(3) In this paper, we designed a probabilistic adaptive fusion strategy based on Bayesian decision theory, which realized the dynamic weight tuning of each modal feature and solved the problem of weight fixing in existing research.

2. Methods

2.1. Data preprocessing

First, to remove noise and interference without distorting the temporal characteristics of the signals, a comprehensive preprocessing pipeline was applied. Zero-phase filtering (implemented via forward-backward processing) was employed for all filtering operations. Specifically, 2nd-order Butterworth bandpass filters were used with frequency bands of 0.5–60 Hz for ECG and 20–600 Hz for PCG signals, respectively. This effectively eliminated baseline wander and high-frequency noise. Furthermore, 50 Hz power line interference was removed using 2nd-order Butterworth notch filters, applied with the same zero-phase technique to prevent phase shifts. After filtering, all signals were standardized using Z-score normalization to ensure uniform feature scaling. To enhance the analytical capability for heart rhythm and sequence patterns, the processed signals were segmented into 5-s epochs using a sliding window with a 3-s step size to generate samples for model input. The 5-s window duration ensures capture of sufficient consecutive cardiac cycles (5–8 cycles at typical heart rates) essential for arrhythmia detection, while the 3-s step size provides 40% overlap to maintain continuity in detecting transitional cardiac events.

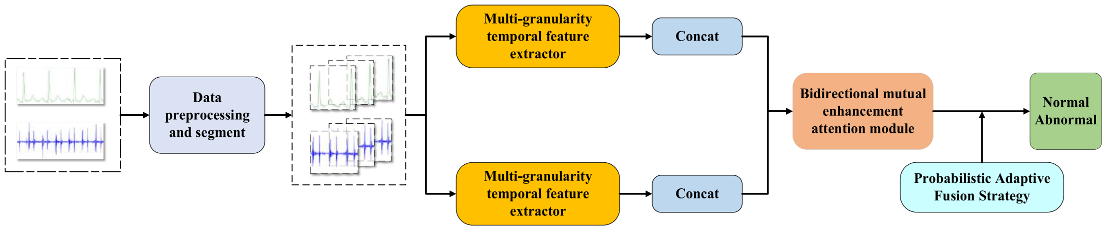

Figure 1. Multi-modality analysis method framework.

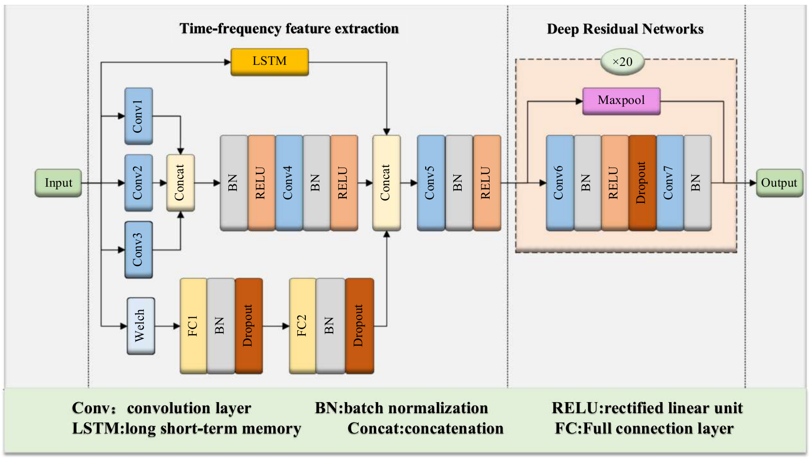

Figure 2. Dual-path multi-scale residual network structure.

2.2. Network design

The framework of the proposed multi-modality CVD analysis method is depicted in figure 1. First, synchronously acquired ECG and PCG signals were preprocessed and segmented into paired fragments. These processed signals were then fed into a dual-path multiscale residual network to extract time-frequency domain fusion features for ECG and PCG, respectively. Subsequently, the fused features were passed through the bidirectional mutual enhancement attention module to exploit complementary inter-modal dependencies. Finally, the probabilistic adaptive fusion strategy dynamically adjusted the weights of ECG and PCG modalities based on their confidence scores, CVD classification.

2.2.1. Dual-path multiscale residual network

For feature extraction, the ECG and PCG signals were fed into the dual-path multiscale residual network structure shown in figure 2, respectively. The time-frequency feature extraction module in this network structure has the ability to extract the multiscale features of the time-domain signals as well as the power spectrum information of the frequency-domain signals. In time domain feature extraction, in this paper, the long-range temporal dependence of ECG and PCG signals was firstly captured by the long short-term memory (LSTM) network (Zhou *et al* 2020) (Schmidhuber and Hochreiter 1997), and then the multiscale local features were extracted by three parallel convolutional layers with different kernel sizes (K4, K16, and K32) in order to comprehensively capture the different-scale waveform features in the electrocardiography and heart sound signals. Multi-scale feature extraction process can be expressed as:

$$F_{multi-scale} = concat(F_{K4}, F_{K16}, F_{K32}) \tag{1}$$

For frequency domain feature extraction, the preprocessed signals were subjected to Welch's method LI *et al* (2023) to compute the power spectral density (PSD) with a window size of 256 and an overlap rate of 50% to obtain the frequency domain features and learn the abstract representations through a two-layer fully connected network (128-128). The specific calculation process of Welch's method can be expressed as:

$$PSD(M, f) = \frac{1}{N_w \cdot P_s} \sum_{a=1}^{A} |X_a(M, f)|^2 \tag{2}$$

Where, $M \in (ECG, PCG)$, N_w represents the number of windows, P_s denotes the power coefficient of each window, a represents the segment number, and $X_a(f)$ represents the Fourier transform of the a-th segment.

The time-domain and frequency-domain features were then input into the deep residual network for further extraction and fusion by channel splicing. The deep residual module consists of 20 residual blocks

Table 1. Parameter table.

Module		Layer	Parameter	Layer	Parameter
Time-frequency feature extraction module		Conv1	K4,S1,C16	Conv2	K16,S1,C16
		Conv3	K32,S1,C16	Conv4	K16,S1,C32
		Conv5	K16,S1,C32	LSTM	U32
		FC1	U128	FC2	128
Deep residual module	Res-1/ Res-2	Conv6	K16,S1,C32	Conv7	K16,S1,C32
	Res-3/ Res-4	Conv6	K16,S1,C32	Conv7	K16,S1,C32
	Res-5/ Res-6	Conv6	K16,S1,C32	Conv7	K16,S1,C32
	Res-7/ Res-8	Conv6	K16,S1,C64	Conv7	K16,S1,C64
	Res-9/ Res-10	Conv6	K16,S1,C64	Conv7	K16,S1,C64
	Res-11/ Res-12	Conv6	K16,S1,C64	Conv7	K16,S1,C64
	Res-13/ Res-14	Conv6	K16,S1,C128	Conv7	K16,S1,C128
	Res-15/ Res-16	Conv6	K16,S1,C128	Conv7	K16,S1,C128
	Res-17/ Res-18	Conv6	K16,S1,C256	Conv7	K16,S1,C256
	Res-19/ Res-20	Conv6	K16,S1,C256	Conv7	K16,S1,C256

designed to extract the fused deep features. Each residual block consists of two convolutional layers and a maximum pooling layer, where the convolutional layer is responsible for feature extraction and the maximum pooling layer serves to reduce the feature size. In the whole network structure, BN layer, RELU layer, and Dropout layer are introduced to improve the training speed of the model, mitigate the gradient vanishing problem of backpropagation, and prevent training overfitting, respectively.

The dual-path multiscale residual network and the input sequences are defined in this paper as $\varphi(\mathbf{x}_M, \theta_M)$ and $\mathbf{x}_M \in \mathbb{R}^{C\times T}$, respectively, where $M \in (ECG, PCG)$, θ denotes the parameters. The two sequences are input the two sequences into the dual-path multiscale residual network. The output result after time-frequency feature extraction module and depth residual module could be expressed as:

$$feature\_out = \varphi(\mathbf{x}_{ECG}, \theta_{ECG}) + \varphi(\mathbf{x}_{PCG}, \theta_{PCG}) \quad (3)$$

Table 1 details the structural parameters of the dual-path multiscale residual network. Where K is the convolution kernel size, S is the step size, C is the number of output channels, U is the number of hidden units, the maximum pooling step is set to 2, the dropout rate is set to 0.3, and finally the fused result is output.

2.2.2. Bidirectional mutual enhancement attention module

To address the inconsistent sampling rates of ECG (1000 Hz) and PCG (2000 Hz), the PCG signal was downsampled to 1000 Hz to ensure temporal alignment. A block alignment strategy was adopted to reduce computational complexity, where time-frequency fusion features of ECG and PCG served as input (Zhang *et al* 2024b). The feature dimensions of ECG and PCG were both denoted as B×C×H, where B represented the training batch size, C was the number of output channels of the two-path multiscale residual network 256, and H represented the length of the ECG and PCG signal sequences 2000, and in order to reduce the computational resources, the signal sequences were divided into 20 blocks, denoted as '**a**', and the length of the sequence of each block was 100, denoted as '**b**'. The ECG blocks and PCG blocks that were correspondent to each other were input into the bidirectional mutual enhancement attention module (Gheini *et al* 2021), and the features of the two modalities were encoded and processed separately for different linear transformation matrices.Next, the interaction weights between ECG and PCG features were calculated and the obtained weights were normalized by applying the *softmax* function. The features of one modality were made more focused on the key information in the other modality to realize the complementary enhancement between the two modalities. The processed 20 groups of blocks were dimensionally spliced and converted to the original format B×C×H. The overall flow of the model is shown in figure 3. For ease of presentation, the following was an example of the feature enhancement process of ECG (PCG is the same), and the specific computational process can be expressed as follows:

$$Q_{PCG} = \boldsymbol{\Phi}_{PCG} \times \boldsymbol{W}_{\mathbf{Q}_{PCG}} \quad (4)$$

$$\boldsymbol{K}_{ECG} = \boldsymbol{\Phi}_{ECG} \times \boldsymbol{W}_{\boldsymbol{K}_{ECG}} \quad (5)$$

$$V_{ECG} = \boldsymbol{\Phi}_{ECG} \times W_{V_{ECG}} \quad (6)$$

$$\Phi_{ECG}^{att} = softmax\left(\frac{\mathbf{Q}_{PCG}\boldsymbol{K}_{ECG}^{T}}{\sqrt{d_K}}\right)\mathbf{V}_{ECG} \quad (7)$$

Where $\boldsymbol{\Phi}$ denotes the input feature sequence, $\mathbf{Q}_{PCG}$ denotes the query, and K_{ECG} is utilized to denote the key to obtain the attention score by performing dot product operations on $\mathbf{Q}_{PCG}$ and $\boldsymbol{K}_{ECG}$.

Subsequently, the attention scores are normalized and probabilized by *softmax* activation function. In order to prevent the parameters from being updated properly because the gradient of *softmax* is too small, and the result of dot product operation on $\mathbf{Q}_{PCG}$ and $\mathbf{K}_{ECG}$ is scaled according to the ratio of $1/\sqrt{d_k}$, and

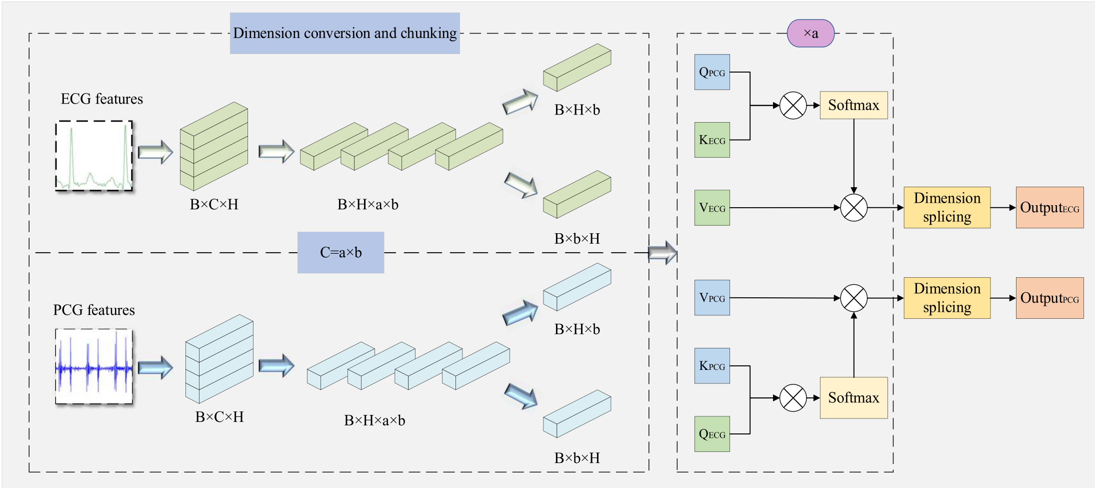

Figure 3. Bidirectional mutual Enhancement attention module.

finally each token in $\mathbf{V}_{ECG}$ is assigned a weight according to the attention score matrix.

2.2.3. Adaptive fusion strategy

To dynamically optimize the weight allocation of ECG and PCG multi-modality information, this paper proposed a probabilistic adaptive fusion strategy based on Bayesian decision theory Mcnamara, Chen (2022). The strategy is based on the following core assumptions: the classification confidence of each modality for a disease should be positively correlated with its contribution, negatively correlated with its loss function, and dynamically adjusted with sample characteristics.

First, the enhanced processed ECG and PCG feature vectors are input to the confidence assessment network, which adopts a three-layer fully-connected layer (128-64-1) structure, and the outputs are normalized to the [0,1] interval by the Sigmoid function, which obtains the corresponding confidence values C_M, where $M \in \{\text{ECG, PCG}\}$, and guarantees $C_M \in [0.01,\ 0.99]$,which defines the joint confidence level:

$$Jo\ \mathrm{int}\ \_confidence = C_{ECG} \times C_{PCG} \tag{8}$$

In order to optimize the modal weight assignment, the holographic factor M-holo is introduced in this study, $M \in \{\text{ECG},\ \text{PCG}\}$ computerized and its mathematical expression is as follows:

$$ECG\_holo = \frac{\log(1 + C_{PCG})}{\log(1 + Jo\ \mathrm{int}\ \_confidence)} \tag{9}$$

$$PCG\_holo = \frac{\log(1 + C_{ECG})}{\log(1 + Jo\ \mathrm{int}\ \_confidence)} \tag{10}$$

This design is based on the principle of mutual information in information theory, so that the weight coefficient of a modality not only depends on its own confidence, but is also affected by the confidence of the opposing modality. When the confidence of one modality is high, the holographic factor would adaptively increase the weight of the other modality(Cao *et al* 2024). Finally, the confidence values are combined with the holographic factors to obtain the combined confidence:

$$CB_{ECG} = ECG\_holo + C_{\text{ECG}} \tag{11}$$

$$CB_{PCG} = PCG\_holo + C_{\text{PCG}} \tag{12}$$

The respective combined confidence levels are obtained through *softmax* normalization, denoted as λ_M, where $M \in \{\text{ECG},\ \text{PCG}\}$, and the fusion process integrates the classification results of each modality by weighting the weights in a weighted manner, which achieves the probabilistic adaptive fusion based on the confidence level.

2.3. Experimental design

2.3.1. Dataset

The study utilized the 2016 PhysioNet/CinC Challenge public dataset as the experimental benchmark, selecting the training-a subset as the core analytical sample(Liu *et al* 2016). This subset consisted of 405 original records with ECG and PCG signals sampled at 1000 Hz and 2000 Hz, respectively, comprising 117 control group participants and 288 CVD patients. To ensure data quality and reliability of subsequent analysis, rigorous preprocessing was performed on the initial dataset of 409 patient groups. Specifically, 17 groups of samples severely contaminated by noise were excluded based on predefined signal quality thresholds, resulting in 388 patients for final analysis (271 CVD patients and 117 control group participants).The dataset was split at the patient level using an 80/10/10 ratio for training, validation, and testing respectively to prevent data leakage. This patient-level splitting strategy ensured that no individual patient's data appeared across multiple splits, maintaining the integrity of model evaluation. The resulting distribution allocated approximately 310 patients to training

(217 CVD, 93 control group participants), 39 patients to validation (27 CVD, 12 control group participants), and 39 patients to testing (27 CVD, 12 control group participants).

To address the significant class imbalance between control group participants and CVD patients, we implemented a comprehensive data augmentation strategy exclusively for the minority class (control group participants) and applied only to the training set. Our augmentation protocol included four techniques: Gaussian noise addition (with standard deviation of 5% of the original signal's standard deviation), time shifting (5%–15% of signal length), amplitude scaling (0.8–1.2× range), and signal inversion (50% temporal reversal, 50% amplitude inversion). Each control group participant in the training set generated exactly one augmented sample, creating approximately 93 additional samples and effectively improving the class balance from the original 93:217 ratio to approximately 186:217 in the final training distribution.

2.3.2. Model training and testing

The TF-CrossNet multi-modality classification network proposed in this study was implemented on a virtual GPU high-performance computing platform based on a 12-core Intel Xeon Platinum 8352V processor (base frequency of 2.10 GHz, maximum RWI of 3.5 GHz) and 32 GB of graphic memory, with a development environment of Pytorch 1.11.0 deep learning framework (Python 3.8, CUDA 11.3) running on Ubuntu 20.04 operating system, using AdamW optimizer (weight decay coefficient $\lambda = $ 1e-4), cosine annealing preheating learning rate strategy (learning rate $\eta = 0.0005$, minimum learning rate η_min = 1e-6), Dropout regularization of 0.3, L2 weight of 1e-5 decay, a fixed batch size of 128, and an early stopping mechanism by improving the loss function by less than 1e-4 in 15 consecutive training cycles, while setting the random number seed to 42.

2.3.3. Evaluation metrics

To comprehensively evaluate the performance of TF-CrossNet, five standard evaluation metrics were employed: accuracy, sensitivity, specificity, F1 score, and the area under the receiver operating characteristic (ROC) curve (AUC). Additionally, to assess noise robustness, we introduced the robustness index (RI) metric. The mathematical definitions are as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{13}$$

$$Sensitivity = \frac{TP}{TP + FN} \tag{14}$$

$$Specificity = \frac{TN}{TN + FP} \tag{15}$$

$$F1 = \frac{2TP}{2TP + FP + FN} \tag{16}$$

Table 2. Comparison of unimodal and multi-modality classification results.

	Acc(%)	Sen(%)	Spe(%)	F1	AUC
ECG	79.71	83.81	79.73	0.857	0.85
PCG	81.90	86.20	76.79	0.867	0.87
ECG+PCG	93.13	92.59	97.7	0.930	0.98

$$RI = \frac{Acc_{noisy}}{Acc_{clean}} \times 100\% \tag{17}$$

Where TP, FN, FP, and TN represent the number of true positive, false negative, false positive, and true negative samples, respectively. AUC represents the area under the receiver operating characteristic (ROC) curve, providing a threshold-independent measure of classification performance. The robustness index (RI) quantifies the model's ability to maintain performance under noise conditions by comparing accuracy in noisy environments to clean signal performance.

3. Results

To verify the effectiveness of the proposed method, this paper conducts experiments on a public dataset. In section 3.1, the paper provides a detailed analysis of the importance of time and frequency domain features and the issue of the contribution of ECG and PCG signals. In section 3.2, the proposed method is systematically compared and analyzed in depth with the existing techniques, aiming to clearly point out the performance improvements and advantages of the method in this paper. Finally, in section 3.3, the paper validates the effectiveness of the developed component through ablation experiments.

3.1. Analysis of the contribution of each modal feature to different samples

To evaluate the contributions of ECG and PCG signals to CVD classification, table 2 presents a comprehensive comparative analysis of single-modal and multi-modal classification results. The findings reveals notable performance differences across signal modalities: classification accuracy based solely on ECG signals was approximately 79.71%, while PCG signals demonstrate a slightly higher performance at 81.90%. Notably, when ECG and PCG signals are jointly integrated, the classification accuracy significantly improves to 93.13%, demonstrating the potential superiority of multi-modal signal fusion in enhancing CVD diagnostic precision.

The comparative performance of the model across control group participants and patient group participants, as presented in table 3, elucidates the algorithm's discriminative capability across different sample categories. For normal specimens, the model demonstrates exceptionally high performance

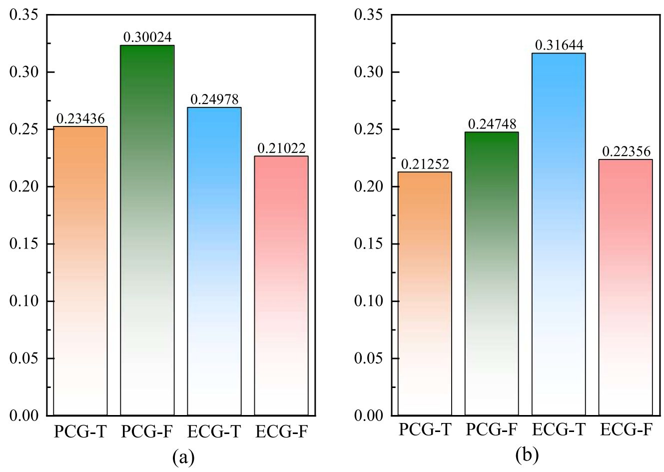

Figure 4. Feature contribution analysis. (a) shows the feature contribution graph of the control group participants, (b) shows the feature contribution graph of the patient group participants.

Table 3. Comparison of classification performance between control group participants and patient group participants.

	Acc	Sen	F1
Normal	0.95	0.90	0.90
Abnormal	0.92	0.98	0.95

metrics, with an accuracy (Acc) of 0.95, sensitivity (Sen) of 0.90, and an F1 score of 0.90. Notably, in pathological samples, despite a marginal reduction in performance indicators, the model maintains robust classification capabilities, achieving a 0.92 accuracy, 0.98 sensitivity, and 0.95 F1 score. These results underscore the model's consistent and reliable diagnostic potential across varied sample conditions.

As shown in figure 4, the results of feature contribution analysis indicate that the PCG frequency domain feature has the highest contribution value (0.30024) in the control group samples, while the ECG time domain feature has the most significant contribution (0.31644) in the patient group samples. This distribution revealed the complementary roles of ECG and PCG features in different categories of samples: the PCG frequency-domain features have a higher sensitivity to the differentiation of health status, while the ECG time-domain features show a stronger discriminative power in disease detection. Therefore, this paper proposed a probabilistic adaptive fusion strategy to optimize the integration process of multi-modality information by dynamically adjusting the weight ratio of ECG and PCG. The experimental results show that compared with the traditional equal-weight fusion strategy, the classification accuracy of this strategy on the test set is improved by 1.08%, which verifies the effectiveness of the dynamic weight assignment in improving the classification performance of the model.

3.2. Comparative experiment

To ensure rigorous experimental fairness, all baseline methods (Chakir *et al* 2020, Li *et al* 2021, 2022, Zhu *et al* 2025) were retrained and optimized using exactly the same preprocessing protocols and identical computational environments as TF-CrossNet.The experimental results and confusion matrix comparison of the proposed TF-CrossNet model against existing methodologies on the PhysioNet/CinC challenge dataset are illustrated in table 4 and figure 5 The Area Under the Curve (AUC) performance comparison between existing approaches and the current model is depicted in figure 6. The comprehensive analysis revealed that the proposed TF-CrossNet model achieved remarkable performance improvements in CVD detection. Specifically, the model demonstrated outstanding classification metrics, including an accuracy of 93.13%, sensitivity of 92.59%, specificity of 97.7%, and an F1 score of 0.93. Notably, this represents the highest accuracy improvement of 0.63% among comparative methodologies. These empirical findings conclusively substantiate the superior performance and exceptional efficacy of the TF-CrossNet model in CVD classification tasks.

In a comparative analysis, In 2020, Fatima Chakir *et al* extracted multiple biomarkers based on synchronized ECG and PCG signals, demonstrating the potential of multi-modality fusion in CVD detection. However, the method required complex manual feature extraction and failed to fully quantify the diagnostic weights of the two modalities in different diseases, which limited its diagnostic efficacy. In contrast, the TF-CrossNet model proposed in this paper can realize end-to-end CVD detection without manually designing features, automatically mining key features in ECG and PCG signals through a dual-path multi-scale residual network and bidirectional mutual enhancement attention module, and dynamically

adjusting the fusion weights between modalities by using the probabilistic adaptive fusion strategy,

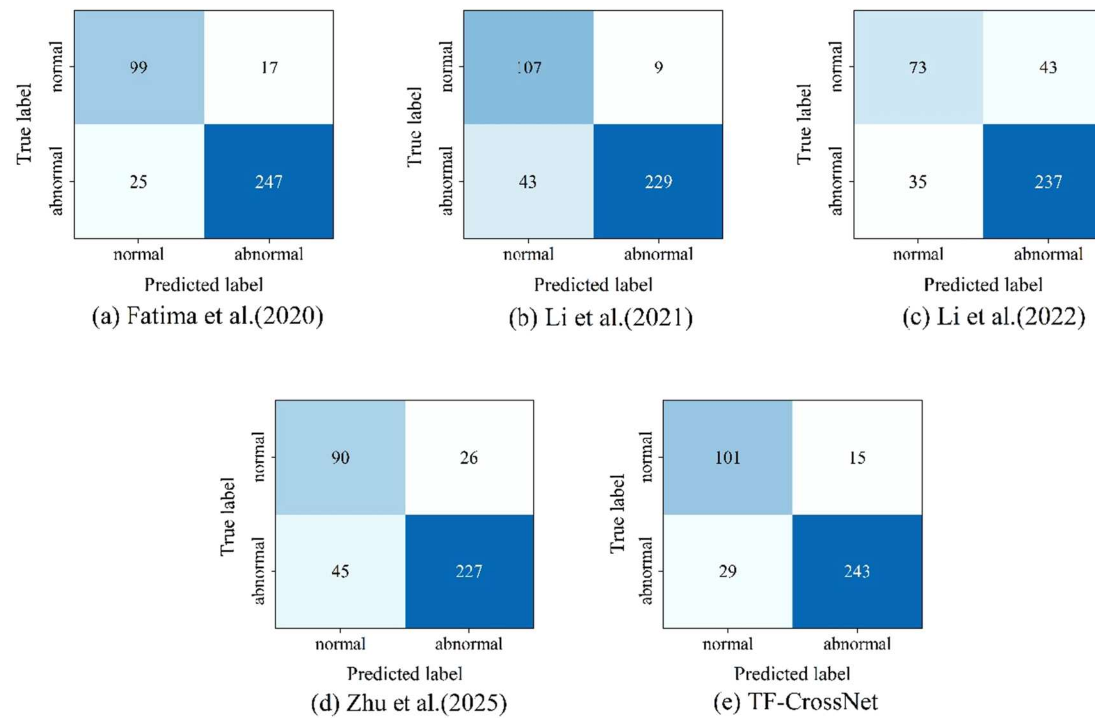

Figure 5. Confusion matrix comparison.

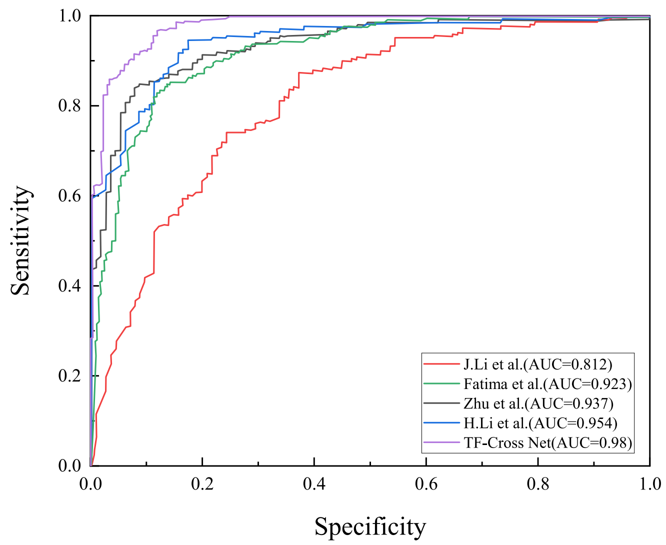

Figure 6. AUC curve comparison.

Table 4. Compare experimental results.

Method	Acc(%)	Sen(%)	Spe(%)	F1	AUC	Interpretability
Chakir *et al* (2020)	92.5	92.31	92.86	0.906	0.95	Low
Li *et al* (2021)	92.5	**95.6**	89.4	0.925	0.946	Medium
LI *et al* (2022)	86.42	84.96	**98.26**	0.911	0.931	Low
Zhu *et al* (2025)	91.6	92.0	91.1	0.915	0.962	Medium
TF-CrossNet	**93.13**	92.59	97.7	**0.930**	**0.98**	**High**

which resulted in a more accurate CVD detection.2021 The studies by Li Han *et al* in 2021 and Li Jinghui *et al.* in 2022 did not fully consider the intermodal weights in their models, although they used deep learning methods to achieve end-to-end ECG and PCG feature extraction. For cardiovascular

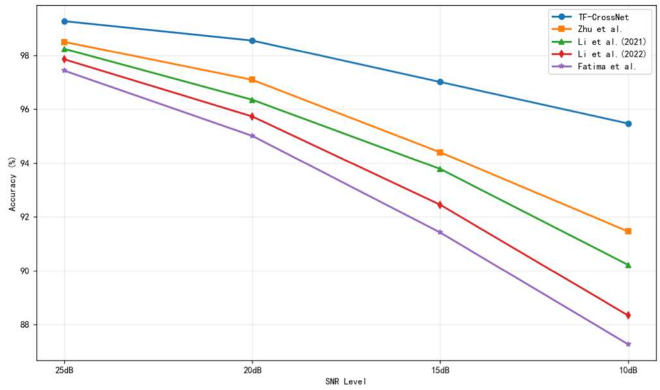

Figure 7. Robustness index analysis.

diseases, the two modalities did not have the same contribution in disease classification. In contrast, the TF-CrossNet model innovatively introduces a probabilistic adaptive fusion strategy, which can adaptively decide the intermodal weight assignment according to the classification of individual modalities, thus accurately integrating information from different modalities and improving the classification performance.2023 Zhu Jiayuan *et al* used a deep learning method for cardiovascular disease classification, but the method was not sufficiently efficient in processing the PCG and ECG signals failed to fully exploit their inherent physiological associations and temporal dependencies. The TF-CrossNet model, however, captures the inherent dynamic interaction patterns between modalities through the bidirectional mutual enhancement attention module, enhances the feature representation, and improves the overall discriminative ability of the model by assigning weights to different modalities through the probabilistic adaptive fusion strategy. The TF-CrossNet model outperforms the existing methods in several evaluation metrics, which is attributed to the dual-path multiscale residual network, bidirectional mutual enhancement attention module and probabilistic adaptive fusion strategy, which enable the model to extract and fuse key features in ECG and PCG signals in a more comprehensive way.

In order to comprehensively evaluate the applicability of the TF CrossNet model in practical clinical environments, this paper conducted noise robustness experiments. In clinical practice, ECG and PCG signals are often affected by various environmental noise, equipment noise, and physiological interference. Therefore, the noise robustness of the model is an important indicator for evaluating its practicality.

In the experimental design, we manually added Gaussian white noise of different intensities to the original test dataset, with noise intensities set at four signal-to-noise ratio levels of 25dB, 20dB, 15dB, and 10dB. These noise levels cover actual clinical scenarios ranging from mild interference (25dB) to severe noise pollution (10dB). All comparison methods were tested under the same noise conditions to ensure fairness and comparability of the experiment.

Table 5. Performance comparison under white noise conditions.

Method	Clean	25dB	20dB	15dB	10dB
Chakir *et al* (2020)	92.50	90.12	87.89	84.56	80.73
Li *et al* (2021)	92.50	90.87	89.12	86.73	83.45
Li *et al* (2022)	86.42	84.56	82.73	79.87	76.34
Zhu *et al* (2025)	91.60	90.23	88.91	86.45	83.78
TF-CrossNet	93.13	92.45	91.78	90.34	88.92

Table 5 shows the performance of each method under different noise conditions. The experimental results show that as the noise level increases, the classification accuracy of all methods shows a downward trend, but the TF CrossNet model exhibits the best noise robustness performance.

In order to more intuitively evaluate the noise robustness of each method, we calculated the robustness index. Figure 7 shows the robustness index analysis results of each method. TF CrossNet maintains the highest robustness index at all noise levels, with an average robustness index of 94.20%, significantly better than the average values of other comparison methods.

3.3. Ablation experiment

To evaluate the contribution and impact of each component of the TF-CrossNet model on the overall performance, a series of ablation experiments are designed in this paper. Table 6 and figure 8 present the comparative results of each experimental configuration, including the following four configurations:

(1) DMRN: Only dual-path multiscale residual network;

(2) DMRN + BMEA: Dual-path multiscale residual network and bidirectional mutual enhancement attention module;

(3) DMRN +AFS: Dual-path multiscale residual network and adaptive fusion strategy;

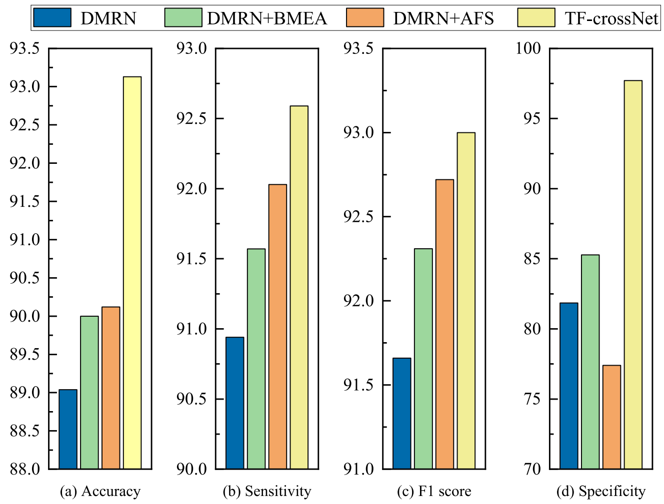

Figure 8. Ablation experiment results.

Table 6. Ablation experiment results.

Model	Acc(%)	Sen(%)	F1(%)	Spe(%)
DMRN	89.04	90.94	91.66	81.85
DMRN + BMEA	90.00	91.57	92.31	85.27
DMRN + AFS	90.12	92.03	92.72	77.4
TF-CrossNet	93.13	92.59	93.00	97.7

(4) TF-CrossNet: It consists of a dual-path multiscale residual network, bidirectional mutual enhancement attention module and adaptive fusion strategy .

Experimental results show that the dual-path multiscale residual network achieves 89.04% accuracy, 90.94% sensitivity, 91.66% F1 score, and 81.85% specificity on the CVD diagnostic task, providing an effective baseline for subsequent improvement. After the introduction of the bidirectional mutual enhancement attention module, the model performance is significantly improved: accuracy increased to 90.00%, sensitivity to 91.57%, F1 score to 92.31%, and specificity to 85.27%. When the dual-path multiscale residual network integrated the probabilistic adaptive fusion strategy, the accuracy increased to 90.12%, the sensitivity reached 92.03%, and the F1 score increased to 92.72%. The model specificity decreased to 77.4%, which was 4.45% lower than the baseline. This suggests that the probabilistic adaptive fusion strategy improves the model's ability to recognize patient group samples while at the same time having an impact on the classification accuracy of control group samples. This trade-off may stem from the model's dynamic weight allocation mechanism for different modal features, which differentially affects the recognition of specific categories in the process of optimizing the overall performance.

Table 7. Top-5 hyperparameter configurations from the random search.

Batch_size	Dropout	Lr	Acc(%)	Time
32	0.5	0.001	93.49	29.17m
128	0.3	0.0005	93.13	13.86m
64	0.5	0.0005	93.07	39.04m
64	0.3	0.0008	93.01	10.78m
128	0.2	0.0005	92.77	14.74m

When the bidirectional mutual enhancement attention module and the probabilistic adaptive fusion strategy are applied simultaneously, the performance of the TF-CrossNet model reaches the optimal state: the accuracy was significantly improved to 93.13%, which was 4.09% higher than the baseline; the sensitivity reached 92.59%, the F1 score was increased to 93.00%, and the specificity was substantially increased to 97.7%, which was 15.85% higher than the baseline. This result fully proves that the proposed two core modules have significant synergistic effects, which can achieve efficient intermodal information interaction at the feature level, while optimizing modal weight allocation at the decision level, thus improving the classification ability of the model.

In summary, the results of the ablation experiments validate the effectiveness of the components of the TF-CrossNet model and their synergistic effects. The bidirectional mutual enhancement attention module enhances the representation capability of the model by facilitating cross-modal feature interactions, while the probabilistic adaptive fusion strategy optimizes the process of integrating multi-modality information through the dynamic weight allocation mechanism. The complementary effects of these two modules enable TF-CrossNet to enhance the overall performance of the CVD detection task.

Table 8. Ablation study on network depth and LSTM module.

Model	Parameter	Acc(%)	Sen(%)	Spe(%)	F1 score(%)
residual network×5	1670297	90.41	93.18	92.29	93.21
residual network×10	1706777	91.33	94.62	94.22	93.41
residual network×15	2168217	92.01	95.73	94.97	94.68
residual network×20	2497817	93.13	98.14	96.70	94.80
residual network×25	3088345	90.48	91.28	94.00	92.57
Without LSTM	1901721	90.53	91.33	91.79	92.47

3.4. Model selection and hyperparameter analysis

To ensure the robustness and optimal performance of our proposed TF-CrossNet model, we conducted an extensive hyperparameter sensitivity analysis using a random search strategy over 20 distinct configurations. This analysis aimed to evaluate the model's performance sensitivity to key hyperparameters including batch size, dropout rate, and learning rate. The primary objective was to identify a configuration that balanced high classification accuracy with computational efficiency, ensuring practical applicability.

As summarized in table 7, the top-five performing configurations all achieved validation accuracy above 93%, demonstrating the robustness of TF-CrossNet to hyperparameter variations. Notably, the configuration with batch_size = 32, dropout = 0.5, and learning_rate = 0.001 yielded the highest accuracy of 93.49%. However, considering the significant computational overhead, we prioritized efficiency for practical deployment. Consequently, the configuration with batch_size = 128, dropout = 0.3, and learning_rate = 0.0005 was selected as the optimal setup for all subsequent experiments. This configuration achieved a highly competitive accuracy of 93.13% with a substantially reduced training time of 13.86 min, representing an optimal trade-off between performance and computational cost.

In addition to the ablation of core architectural components, we further investigated the impact of model depth and critical sub-modules on performance. Specifically, we evaluated our dual-path multiscale residual network with varying depths and examined the contribution of the LSTM module for capturing long-term temporal dependencies.

The results presented in table 8, indicate that model performance improved with increasing depth up to 20 residual layers, achieving optimal metrics. Further increasing the depth to 25 layers led to a performance decline , suggesting potential overfitting or optimization difficulties. This confirms that our chosen 20-layer depth is near-optimal for this task. Furthermore, removing the LSTM module resulted in a significant performance drop across all metrics, unequivocally justifying its inclusion for capturing long-range temporal dependencies in our final architecture.

4. Discussion

The experimental results demonstrate that TF-CrossNet achieved superior performance in cardiovascular disease classification, with 93.13% accuracy, 97.7% specificity, and 0.98 AUC.

The feature contribution analysis revealed distinct patterns between control group participants and patient group participants, with PCG frequency-domain features showing highest contribution for control group samples identification (0.30024) while ECG time-domain features dominated disease discrimination (0.31644). This differential contribution pattern can be attributed to the physiological nature of cardiac function: normal mechanical activity produces consistent acoustic signatures in the frequency domain, while pathological electrical abnormalities manifest more prominently in ECG temporal patterns. This finding validates our hypothesis that ECG and PCG modalities provide complementary diagnostic information. The ablation study results provide insights into the contribution of individual components. The baseline dual-path multiscale residual network achieved 89.04% accuracy, with bidirectional mutual enhancement attention contributing 0.96% improvement and adaptive fusion strategy adding 1.08%. The synergistic effect of combining both components (final accuracy: 93.13%) demonstrates that cross-modal feature interaction and dynamic fusion address different aspects of multi-modal learning. The particularly strong specificity (97.7%) indicates robust false positive control, which is clinically significant for reducing unnecessary patient anxiety and testing burden. The 0.63% accuracy improvement over existing methods, while appearing modest, represents meaningful progress in a mature field where performance gains become increasingly challenging. Our superior AUC performance (0.98) compared to existing methods indicates improved discriminative capability across different decision thresholds, suggesting better feature representation separability between control group participants and patient group participants. The enhanced interpretability through adaptive weight visualization provides clinical advantages that traditional black-box approaches cannot offer.

The noise robustness experiments provide critical insights into TF-CrossNet's clinical applicability under realistic conditions. Our model demonstrated exceptional resilience to noise interference, maintaining 88.92% accuracy even under severe 10dB SNR conditions while achieving an average robustness index of 94.20% across all tested noise levels. This superior noise tolerance significantly outperformed existing methods, with improvements ranging from 5.14% to 12.58% under the most challenging conditions. The robust performance can be attributed to the multi-scale feature extraction capability that captures signal characteristics across different temporal and frequency scales, making the model less susceptible to noise-induced feature distortion. Additionally, the bidirectional mutual enhancement attention mechanism enables effective noise mitigation through cross-modal information compensation—when one modality is corrupted by noise, the other can provide complementary information to maintain diagnostic accuracy. The adaptive fusion strategy further enhances robustness by dynamically adjusting modality weights based on confidence levels, effectively down-weighting noise-contaminated signals during decision making. These findings are particularly significant for clinical deployment, as real-world ECG and PCG recordings are frequently affected by motion artifacts, electromagnetic interference, and ambient noise. The demonstrated noise robustness suggests that TF-CrossNet could maintain reliable performance in diverse clinical environments, from controlled hospital settings to ambulatory monitoring scenarios.

Several limitations should be acknowledged. First, our study was validated on a single dataset (PhysioNet/CinC 2016), which may limit generalizability across diverse patient populations and clinical settings. Second, the binary classification framework, while establishing proof-of-concept, does not address the spectrum of cardiovascular disease subtypes encountered in clinical practice. Additionally, the computational complexity of our approach may present challenges for real-time deployment in resource-constrained environments. Future research should focus on multi-center validation across diverse patient populations to establish clinical generalizability. Extension to multi-class classification frameworks would better reflect the spectrum of cardiovascular disease presentations and enable more precise risk stratification. Integration of additional physiological modalities such as photoplethysmography and blood pressure variability could further enhance diagnostic accuracy. Computational optimization for real-time deployment and development of edge computing solutions would facilitate widespread clinical adoption. Finally, prospective clinical trials are needed to validate the practical utility and cost-effectiveness of this approach in real-world healthcare settings.

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Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: https://archive.physionet.org/physiobank/database/challenge/2016/.

Ethics statement

This study was performed in accordance with the Declaration of Helsinki. All adult participants provided written informed consent for publishing identifying details in this study.

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